

Photoconductive Semiconductor Switch Experiments for Pulsed Power Applications

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Abstract—Photoconductive semiconductor switches (PCSS's) are being developed at SNL for pulsed power applications that cannot be built with traditional high-power switching technology. For a variety of projects, we have constructed silicon or GaAs switches which have switched (not all at one time) over 120 kV and 4 kA, demonstrated less than 270-ps rise times, contributed a resistance of less than $1\ \Omega$ and a stray inductance of less than 10 nH, produced a burst of 20 bipolar pulses at 20 MHz, been triggered in a high-gain mode (called lock-on) with less than 100 μ J using laser diode arrays operated at 10 Hz, and demonstrated recovery from lock-on in 35 ns. This paper describes the highlights of this research effort and our approach in applying photoconductivity to pulse power switching.

I. INTRODUCTION

IN NORMAL photoconductivity, i.e., at low fields and with near-visible radiation, only one electron-hole pair is created for each photon absorbed in the semiconductor. Switches are produced by using nearly intrinsic, high-resistivity semiconductors and an intense light pulse to lower the resistivity rapidly, but they are relatively inefficient because they have no gain. Nevertheless, these so-called "linear" PCSS's are of special interest for some of their switching properties which surpass high-gain, high-power switches such as spark gaps and thyatrons. Some semiconductors, GaAs and InP, produce PCSS's with an apparent gain at high fields. Once triggered, these nonlinear switches stay on until their field drops below a characteristic value. Subnanosecond rise time, optical triggering, picosecond jitter, and nanosecond recovery make both types of these switches attractive for pulse power applications ranging from wide-bandwidth impulse radar to compact-pulsed accelerators and simulation of subnanosecond rise time electromagnetic pulses from nuclear events. The following sections present results from experiments designed to use both linear and nonlinear PCSS's as efficiently as possible to take advantage of their unique properties in the pulse power regime.

II. SWITCH CONFIGURATIONS

Although we have tested many different types of switch configurations, in most of our schemes a surface of the semiconductor is illuminated which is parallel to the field

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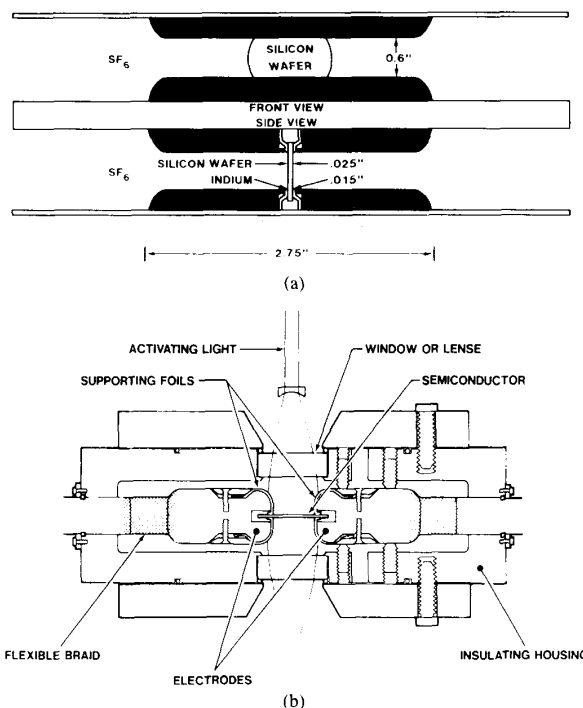


Fig. 1. Photoconductive semiconductor switch configurations. Both use 1-inch-diameter by 0.025-inch-thick semiconductor wafers and produce uniform macroscopic fields near the switch. (a) Clamped configuration which holds the wafer rigidly between parallel plate electrodes. (b) The wafer is suspended by four 0.001-inch-thick copper foils.

sustained by the switch. Two such configurations are shown in Fig. 1. This lateral orientation, as opposed to vertical orientation where the fields are perpendicular to the illuminated surface, has been pursued for the following reasons. First of all, the absorption depth of the activating light is small compared to the length of the semiconductor required to stand-off voltages sufficient for our pulse power applications (10 kV to 5 MV). The absorption depth of above-bandgap, near-visible radiation ranges from a few hundred Angstroms to tens of micrometers depending on the semiconductor and the wavelength of the radiation. On the other hand, the theoretical bulk breakdown strength of semiconductors ranges from 200 kV/cm (GaAs) to a few megavolts per centimeter (SiC and diamond [1]). For even the strongest semiconductors,

several centimeters of semiconductor may be required. If carriers are only created near the surface of the semiconductor, they must drift to the center of a thick switch before the switch resistance is significantly lowered. Even at the saturated drift velocity of approximately 10^7 cm/s, this time of drift may significantly affect the rise time and jitter of large switches.

Another potential problem with carrier drift from the surface of a switch in the vertical orientation is that the field in the center of a thick switch increases as the carriers approach the center. When this field exceeds the bulk breakdown strength of the semiconductor it may lead to impact ionization of sufficient energy to damage the semiconductor and degrade the reliability of the switch. (This effect has been successfully used to initiate avalanche breakdown in small GaAs switches with extremely low light level illumination [2]). Finally, studies of thin-film laminates have shown that the breakdown strengths of these films do not scale linearly with their thickness, but instead thicker dielectrics fail at a significantly lower electric field. To obtain the fastest rise times possible, avoid potential rise time and bulk breakdown difficulties, and develop the simplest scalable PCSS for high-voltage applications, we have concentrated on the lateral configuration. In this case, the field is parallel to an easily illuminated surface and only a thin region of the switch near this surface must be made conductive to activate even large switches. This is not to say that the lateral approach is the only option, nor that it is without problems. The main disadvantage of the lateral configuration, which is discussed in a following section on high-field stress, is the surface flashover problem along the illuminated surface. Several authors are pursuing vertical and other novel configurations [1]–[6].

III. FAST RISE TIME DEMONSTRATION

In order to demonstrate that subnanosecond rise times, which are typical of small low-voltage PCSS, could be achieved with the large lateral PCSS needed for high voltages, a 1-inch-diameter silicon wafer was used to switch the dual stripline shown in Fig. 2. The line was designed to match the size of the switch so that only negligible stray capacitance or inductance would be introduced. The switch was activated with a 100-ps-wide, 3-mJ pulse from a mode-locked Nd:YAG laser. Direct transient waveform measurements (Fig. 3(a)) recorded a 350-ps rise time which was limited by the oscilloscope. Repetitive measurements at 10 Hz (Fig. 3(b)) revealed a 280-ps rise time which was limited by the bandwidth of an electrical delay line [7]. This test was performed in air, so the charge voltage was limited to 10 kV. However, this same size silicon wafer has been used to switch 123 kV in the slower rise time tests described in the following section.

IV. HIGH-FIELD STRESS

The “on” resistance of a PCSS operated in the linear mode when the light pulse is short compared to the carrier

FAST PULSE SWITCH CONFIGURATION

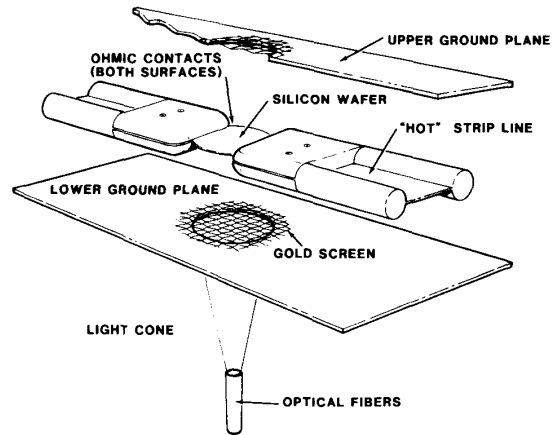


Fig. 2. Apparatus used to demonstrate subnanosecond high-power photoconductive semiconductor switching. This cutaway view shows the 1-inch-diameter wafer held between two pieces of the center line. This dual strip is actually 4 ft long and is completely surrounded by a rectangular ground plane.

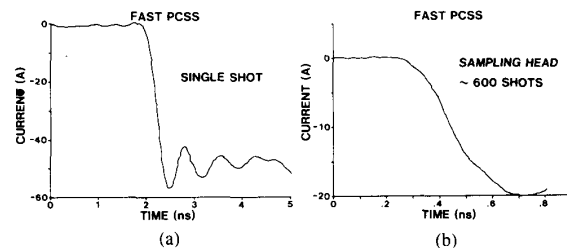


Fig. 3. Fall time measurements for the apparatus shown in Fig. 2. The waveform on the left was recorded from a single shot and the fall time of 350 ps was limited by the oscilloscope. The other waveform is the result of 600 shots at 10 Hz using a sampling head and delay line which limited the rise time to 280 ps [7].

recombination time is given by [8]

$$R_{on} = \frac{L^2 E_{ph}}{E_l e \mu_{eh} (1 - r)} \quad (1)$$

where

L	length of the switch parallel to the field
E_{ph}	energy per photon of the activating light pulse
E_l	total energy in the activating light pulse
e	charge on an electron
$\mu_{eh} = \mu_e + \mu_h$	mobility of electrons and holes
r	optical reflectivity of the sample (~ 0.3).

For a 1.5-cm-long silicon PCSS, a 2.5-mJ laser pulse at $1.06 \mu\text{m}$ will produce an “on” resistance of less than 5Ω . To operate a switch efficiently, it should be as short as possible. This implies that it should be operated under the highest field stress attainable without breaking down

electrically. Consequently, our initial emphasis in PCSS research was to increase the breakdown strength of our large lateral switches (1.5 and 3.3 cm long).

The switch holders which we have tested (Fig. 1) have all been designed to minimize the macroscopic field inhomogeneities and avoid field enhancements which might initiate electrical breakdown. The normal fault mode is flashover from contact to contact along the surface of the wafer that creates a short. We have tested different dielectric media surrounding the switch: 30-psig SF_6 , transformer oil, deionized H_2O , and freon. We have also grown thick oxides on silicon, coated with SiN, bonded to glass, and a variety of other schemes [9]. The highest dark breakdown strengths attained were under deionized water: 143 kV/cm (214 kV) for GaAs and 91 kV/cm (136 kV) for silicon. The best fields reached on samples used for switching were significantly lower, initially around 35 kV/cm. Contacts made with Cr-Cr-Mo-Au [9] have increased dramatically for silicon to 82 kV/cm (123 kV) under pressurized SF_6 . We are still searching for better contacts for GaAs, as the highest field attained while switching our large lateral samples is still only 57.1 kV/cm (85.6 kV). Smaller, 2.0-mm-long, GaAs switches, which were made for lower voltage applications, have withstood fields as high as 70 kV/cm. We have by no means exhausted the possibilities to suppress surface flashover on silicon or GaAs and continue to seek improvements in this area.

In testing our switches at high fields, the current was limited by the impedance of our main test system, which uses a 50- Ω high-voltage transmission line and a variable terminating resistor. The peak current achieved in this apparatus for a 1-in-diameter silicon wafer with 2-cm-wide contacts was 2.0 kA. Tests with a lower impedance transmission line (7 Ω) have achieved 2.8 kA for this same size wafer (1.4-kA/cm contact width). Our smaller GaAs switches have survived linear current densities of 3.2 kA/cm [10].

V. A GAIN MECHANISM: LOCK-ON

Our first experiments with silicon PCSS's were for closing switch applications, because the carrier recombination lifetime in our nearly intrinsic silicon is several hundred microseconds. Once a silicon switch is activated, it stays on until its resistance recovers due to exponential carrier recombination. However, more pressing needs for fast recovery closing and opening (toggling) switches led us to start experiments with Cr:GaAs which typically has a carrier recombination time of only a few nanoseconds. In this case, carriers recombine so rapidly that the switch conductance essentially follows the light pulse. In preliminary tests for a prototype pulsed linear induction accelerator (PLIA), a regenerative RF modulated Nd:YAG laser was used to activate the burst of electrical pulses shown in Fig. 4. This 20-MHz train of bipolar pulses was obtained using a 1.5-cm-long Cr:GaAs PCSS, which was supporting electric fields of less than 8 kV/cm. The shape of the burst followed the shape of the activating light burst

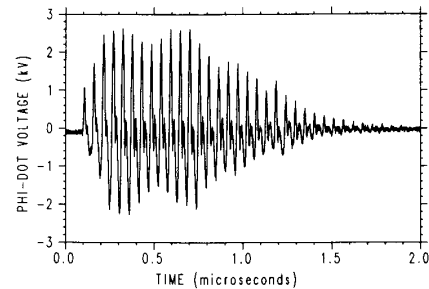


Fig. 4. This burst of pulses was produced using a Cr:GaAs PCSS at low fields to drive an inverting transmission line and a small linear induction accelerator cavity. The phi-dot monitor measures the voltage produced by the cavity on a coil which surrounds the magnetic core. The switch was triggered with an RF-modulated light source.

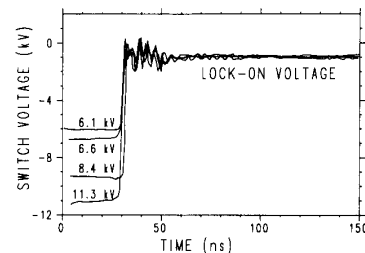


Fig. 5. High-field switching of a 2-mm-long GaAs switch. This plot shows a constant lock-on voltage for various initial charging voltages. The switch was triggered into this lock-on state with a 10-ns-wide laser pulse at 26 ns. It will continue to conduct until the energy is dumped from the circuit and a 5-kV lock-on field can no longer be maintained.

which was 532 nm, the frequency doubled output from the YAG laser. The switch was also used to produce a monopolar, 8-kV pulse train at 40 MHz. A scaled up version of this switch would work nicely for the PLIA except that the total optical energy required to trigger enough switches for the entire accelerator requires an unreasonably large laser. As with silicon, it is essential to operate GaAs at the highest fields possible.

Above 8 kV/cm, a new effect was observed. Fig. 5 shows the results from a Cr:GaAs wafer that was switched after pulse charging to several different voltage levels. The switch was triggered with a single light pulse of similar shape in time to one of the pulses from the light burst described above. At high fields, instead of following the light pulse shape, the switch turned on and stayed on (lock-on) until the energy was completely discharged from the charged transmission line. The voltage across the switch did not go to zero. Instead, it dropped to a constant value, the lock-on voltage, which was independent of the initial charging voltage. The current appeared to be whatever the circuit could supply while maintaining the lock-on voltage drop across the switch [11], [12].

After many tests with different types of GaAs and also one type of InP, we have made the follow observations about lock-on [13], [14]:

- 1) It occurs only when the voltage across the switch is

greater than the lock-on voltage to which the switch settles upon switching.

2) The lock-on voltage of a given switch is given by a material-dependent field times the length of the switch.

3) This field ranges from 3.5 kV/cm on our purest, lowest resistivity GaAs switches to 8 kV/cm on our highest resistivity Cr:GaAs, and was 14.4 kV/cm with InP.

4) The switch must be triggered. As mentioned earlier, we have had a sample hold off 143 kV/cm without switching.

5) The switch can be triggered into lock-on with nearly three orders of magnitude less optical energy than required to reach a comparable resistance at low fields with linear photoconductivity.

This last characteristic of lock-on indicates some type of carrier multiplication mechanism and may allow the use of this type of switch for applications which, due to the optical energy requirement, were previously unattainable. Since the level of the lock-on field is in a range just above the region of the Gunn effect, one explanation for this effect maintains that it is related to the negative differential resistivity exhibited by both GaAs and InP [15]. Multiple Gunn domains might be created which cause large enough field enhancements to produce avalanche carrier generation and explain the gain. Other models suggest that 1) impact ionization of states within the gap, 2) a change in capture cross sections of deep levels at high fields, or 3) a longer recombination time at high fields, or combination of the three effects, might explain the effective gain [13]. To date, a single explanation has not been verified, but experiments to observe Gunn domains and high-field I - V curves are planned, which may help distinguish between the models.

We are presently pursuing the obvious use of lock-on PCSS for closing switch applications ranging from 1 to 200 kV. We have used short (2.0 mm long by 2 cm wide) lateral GaAs switches in low-inductance circuits to switch 4 kA and deliver 3 kV to the load when charged initially to 4 kV. We have triggered narrower (2.5 mm square) lateral GaAs switches with commercially available laser diode arrays. Higher voltage applications for lock-on PCSS are discussed in the next section, but generally lock-on appears to be an efficient optically activated switching scheme which may be useful for many short pulse, closing switch applications.

VI. FAST RECOVERY FROM LOCK-ON

At first glance, one might conclude that the use of GaAs at high fields for closing and opening or toggling applications was ruled out by the lock-on effect. However, since it is feasible to reduce the field across the switch temporarily after switching, it may be possible for the switch to still recover rapidly and withstand high fields after its open-state resistance has returned [15]. The question is, how long must it remain at low fields to recover? To find an answer to this question, additional experiments were performed. Two circuits have been designed and

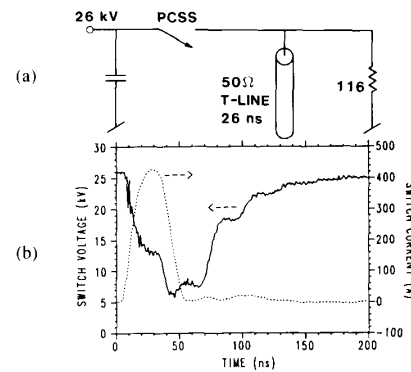


Fig. 6. Recovery from lock-on using a parallel transmission line circuit. The solid line is the voltage across the PCSS (left axis) and the dotted line is the current through the PCSS (right axis). After being triggered into lock-on at 10 ns, the switch sees a reflection from the transmission line at 35 ns which reduces its voltage below lock-on. Recovery of the switch is demonstrated as its current drops to zero by 50 ns. Late time plateaus in the voltage waveform are produced by the intentional mismatch between the load and the transmission line.

tested to induce fast recovery from lock-on, and a third one is being planned.

The first circuit (Fig. 6(a)) uses an open transmission line which is in parallel with the load. Its purpose is to induce carrier recombination in the switch. The line is intentionally undermatched to the load. When the switch closes, it drives the load and charges the transmission line to the charge voltage less the lock-on voltage drop across the switch. Since the transmission line is open ended, it reflects a wave back along the line to the switch and the load. If the switch were out of the circuit, the line would drive the load at a higher voltage than was supplied by the switch because the transmission line has a lower impedance than the load. When this increase in voltage is equal to the original lock-on voltage drop across the switch, the switch will see zero electric field and will have the two-way transit time of the transmission line to recover. Fig. 6(b) shows the voltage and current waveforms measured for a 1.5-cm-long Cr:GaAs switch as it recovers in this configuration. The return to high voltage after the current pulse ended clearly demonstrates recovery. The current pulse lasted about 50 ns and the voltage returned across the switch in 100 ns.

A shorter transmission line could be used to shorten the current pulse and time of recovery, but there are some drawbacks to this circuit which might be avoided. The circuit presumably works best when it is initially charged to the voltage for which the impedance mismatch causes a reflection which exactly zeroes the voltage across the switch. Above or below this optimum initial voltage, the switch will see a non-zero field, which may impede recovery. Secondly, the voltage waveform shows multiple reflections from the transmission line mismatch as the voltage returns across the switch. These reflections are also seen at the load because it is being driven by the transmission line as the switch recovers. Finally, we have found that the switch recovers from lock-on whenever the

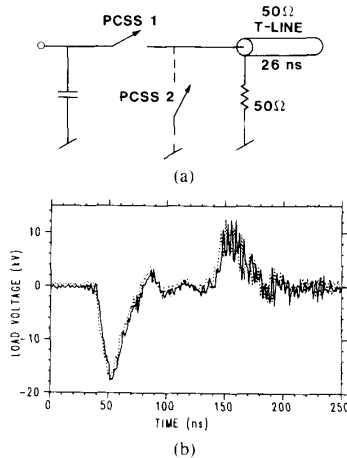


Fig. 7. Recovery from lock-on using a series transmission line circuit. Two waveforms show the reproducibility of this measurement. The first pulse occurs when PCSS 1 is closed and charges the transmission line (through the load). Once the line is charged, the voltage across PCSS 1 is reduced and its resistance returns. The second pulse occurs when PCSS 2 discharges the transmission line.

optical energy used to trigger the switch is greater than 1.5 mJ. Below this value and down to about 500 μ J, it can still be triggered into lock-on, but it does not recover. Instead it stays on until the energy is dumped from the charge capacitor, which takes several microseconds.

To eliminate some of these characteristics, another circuit was designed and tested. In this circuit (Fig. 7(a)), a matched transmission line is used in series with the load. One PCSS is triggered to charge the transmission line, and a second PCSS is used to discharge. When the line is charged, the current in the circuit stops, and the voltage across the charging switch disappears until the discharging switch is triggered. This trigger delay is adjusted to determine the minimum time required for recovery. Since the current through the load changes direction, a bipolar pulse is observed (Fig. 7(b)). Each switch is triggered in the lock-on mode, so lock-on voltage losses from both switches cause the second pulse to be smaller than the first.

We have observed recovery when the discharging switch was triggered as close as 55 ns after the charging switch was triggered. However, even with trigger delays as long as 100 ns, the recovery is not consistent. Triggering and recovery were also observed when the optical trigger energy was reduced to as low as 75 μ J. This is significantly lower than the trigger energy required in the first circuit above, indicating a circuit-dependent effect. Fig. 8 shows the current through the load and the discharging switch for a short burst of pulses where the charging and discharging sequence was repeated at 5 MHz. On the fourth repetition, the discharging switch was triggered before the charging switch had recovered, and since both switches were on, the energy in the charged capacitor was dumped to ground through the switches. This fault mode usually destroys the switches as all of the

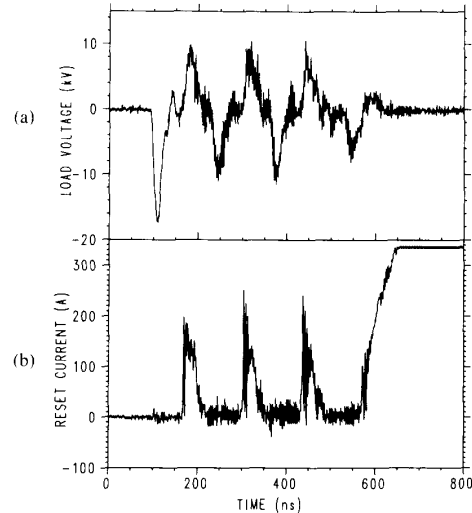


Fig. 8. A short burst of pulses produced by the circuit shown in Fig. 7. Both switches are used to charge and discharge the transmission line repetitively at 5 MHz. (a) Voltage produced across the load. (b) Current through PCSS 2. After three successful repetitions, PCSS 1 fails to recover its resistance before PCSS 2 is triggered, which dumps all the circuit's energy to ground, bypassing the load.

energy in the circuit is dissipated in the switches instead of the load. We are presently trying to determine the factors which influence reliable recovery and will soon be testing a circuit which avoids this destructive fault mode.

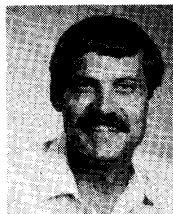
VII. CONCLUSION

We have described our experiments to develop PCSS's for pulsed power applications, which have been built with nontraditional high-power switching technology. A scalable lateral PCSS configuration has been tested which has demonstrated a potential for faster rise times, less jitter, lower inductance, faster recovery and optical triggering for new pulse power projects. We have used these switches as both closing and toggling switches at repetition rates up to 40 MHz. A high-field, gain mechanism (lock-on) was explored and tested which may eliminate the major disadvantage to this type of switch, its requirement for large optical trigger energies.

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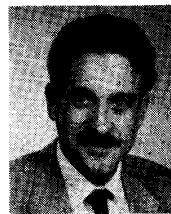


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